

# Scientific Analysis & Research Architecture Report

SUPER MOON — QUALIFIED RESEARCH, AUTOMATION, VERIFICATION & ORCHESTRATION CORE

|                      |                  |                    |                             |
|----------------------|------------------|--------------------|-----------------------------|
| Analysis ID          | 34f9dce09a261638 | Prompt SHA-256     | 34f9dce09a261638725cbf6d... |
| Scientific readiness | 93.3/100         | Architecture score | 93.8/100                    |
| Risk class           | VERY_HIGH        | CRI                | 0.665                       |
| SM32 runtime         | ACTIVE           | Generated          | 2026-08-19 19:43 UTC        |

## Evidence discipline

This document distinguishes prompt-derived analysis from externally validated scientific results. No external solver, laboratory, patent novelty search, certification authority, or third-party qualification is represented as having been executed unless separately evidenced.

# 1. Executive Scientific Assessment

The engine has analyzed prompt structure, scientific governance, risk, reproducibility, verification/validation coverage, UQ, evidence posture and SM32 knowledge alignment. It has not fabricated external solver, laboratory, certification, novelty-search or experimental results.

| Dimension             | Score / 100 |
|-----------------------|-------------|
| Problem Formalization | 100.0       |
| Mathematical Rigor    | 100.0       |
| Verification          | 100.0       |
| Validation            | 66.7        |
| Uncertainty           | 100.0       |
| Reproducibility       | 100.0       |
| Evidence              | 100.0       |
| Orchestration         | 100.0       |
| Risk Governance       | 100.0       |
| Reporting             | 71.4        |

## Primary strengths

- Problem Formalization: 100.0/100
- Mathematical Rigor: 100.0/100
- Verification: 100.0/100
- Uncertainty: 100.0/100

## Primary gaps

- Validation: 66.7/100
- Reporting: 71.4/100
- Problem Formalization: 100.0/100
- Mathematical Rigor: 100.0/100

# 2. Problem Formalization

## Detected objectives

- You are designing the next evolution of SUPER MOON as a high-assurance:
- automate parameter studies, design-space exploration, validation, optimization, and reproducibility; 12.
- CORE DESIGN PHILOSOPHY  
=====
- LEVEL A — INTERNAL COMPUTATIONAL CORE Use SUPER MOON internal implementations for research, prototyping, algorithm development, solver experimentation, comparative studies, low/medium-scale simulations, hypothesis testing, sensitivity analysis, reduced-order models, design-space exploration, surrogate construction, and independent cross-checking.
- Build a formal Computational Risk Index (CRI).
- Create SolverRoutingEngine.

## Constraints and mandatory conditions

- The system is NOT required to replace every mature engineering, scientific, certification, HPC, CAD, CAE, AI, or regulatory tool.

- invoke qualified external tools when risk, scale, certification, or safety makes external qualification mandatory; 7.
- Do not maximize: - number of algorithms; - number of classes; - number of modules; - number of solver names; - number of architecture diagrams; - number of unsupported theoretical claims.
- Weights must be domain-specific rather than universal constants.
- MANDATORY ESCALATION CONDITIONS  
=====
- But internal output alone must NOT be represented as final authority unless qualification evidence explicitly supports that status.
- Qualification must remain use-case-specific.
- Outputs: - recommended internal solver; - recommended qualified external solver; - independent verification solver; - recommended discretization; - expected numerical risks; - validation strategy; - required evidence; - escalation decision; - confidence target.

### Requested outputs

- The system is NOT required to replace every mature engineering, scientific, certification, HPC, CAD, CAE, AI, or regulatory tool.
- formulate complex scientific and engineering problems correctly; 2.
- compare solver outputs; 9.
- C = criticality S = computational scale U = uncertainty I = consequence of incorrect result E = evidence deficiency N = novelty of modeled regime Q = qualification deficiency
- But internal output alone must NOT be represented as final authority unless qualification evidence explicitly supports that status.
- Outputs: - recommended internal solver; - recommended qualified external solver; - independent verification solver; - recommended discretization; - expected numerical risks; - validation strategy; - required evidence; - escalation decision; - confidence target.
- Vertices: variables, fields, equations, constraints, constitutive laws, parameters, solvers, observables, datasets, experiments.
- Report L1, L2, and L-infinity error where appropriate.

### Equation-like statements

- $Quality = f(\text{Correctness}, \text{Verification}, \text{Validation}, \text{Uncertainty},$
- C = criticality
- $CRI = wC*C + wS*S + wU*U + wI*I + wE*E + wN*N + wQ*Q$
- with  $\text{sum}(\text{weights}) = 1$ .
- $0.20 \leq CRI < 0.40$  MODERATE
- $0.40 \leq CRI < 0.60$  HIGH
- $0.60 \leq CRI < 0.80$  VERY\_HIGH
- $CRI \geq 0.80$  CRITICAL
- $\Delta_{AB} = |A-B| / \max(|A|, |B|, \epsilon)$
- Represent  $Y=f(X)$ , with uncertain X.
- Sample  $X_i \sim p(X)$ , compute  $Y_i=f(X_i)$ , and report:
- For residual  $R(u,p)=0$  and objective  $J(u,p)$ , support adjoint workflows.

## 3. Domain & Computational Regime

| Detected domain     | Lexical hits | Confidence |
|---------------------|--------------|------------|
| Scientific Research | 145          | 0.98       |
| HPC / Numerics      | 136          | 0.98       |
| Statistics / UQ     | 81           | 0.98       |
| ML / AI             | 68           | 0.98       |
| Structures / FEA    | 59           | 0.98       |
| CFD / Fluids        | 36           | 0.98       |
| Optimization        | 31           | 0.98       |
| Software / Systems  | 27           | 0.98       |

## 4. Computational Risk & Escalation

| Metric                            | Assessment                     |
|-----------------------------------|--------------------------------|
| CRI                               | 0.6645                         |
| Risk class                        | VERY_HIGH                      |
| Scale class                       | S7_EXTREME_DISTRIBUTED         |
| External qualified tool mandatory | True                           |
| Independent verification required | True                           |
| Human review required             | True                           |
| Engine                            | embedded-supermoon32.qualified |

- Mandatory trigger: airworthiness
- Mandatory trigger: critical\_infrastructure
- Mandatory trigger: extreme\_nonlinear\_structure
- Mandatory trigger: extreme\_sparse\_distributed
- Mandatory trigger: flight\_critical\_structure
- Mandatory trigger: formal\_legal\_acceptance
- Mandatory trigger: human\_safety
- Mandatory trigger: independent\_authority\_review
- Mandatory trigger: industrial\_cad\_interoperability
- Mandatory trigger: life\_critical
- Mandatory trigger: medical\_device\_regulatory
- Mandatory trigger: nuclear\_safety
- Mandatory trigger: production\_gpu\_beyond\_validation
- Mandatory trigger: regulatory\_certification
- Mandatory trigger: very\_large\_production\_cfd

## 5. Verification, Validation and UQ Architecture

Recommended interpretation

- Verification answers whether the equations and algorithms are solved correctly; validation addresses whether the model is adequate for the intended reality or benchmark.
- High-risk conclusions should not rely on a single computational path. Use internal SM32 computation as research/cross-check support and escalate when qualification policy requires it.

- Treat discretization error, parameter uncertainty, model-form uncertainty and data uncertainty as separate budget components.
- Document non-convergence and disagreement as results rather than silently filtering them out.

## 6. Research Orchestration DAG

| Stage | Name                          | Purpose                                                                                                                   |
|-------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 01    | Ingest & normalize            | Hash prompt, identify sections, constraints, claims and domain vocabulary.                                                |
| 02    | Problem formalization         | Define objective, inputs, outputs, governing equations, assumptions, boundary/initial conditions and acceptance criteria. |
| 03    | Dimensional & regime analysis | Check units, nondimensional groups, model validity and operating regime.                                                  |
| 04    | Risk / scale assessment       | Compute CRI and escalation policy. Current inferred class: VERY_HIGH (CRI=0.665).                                         |
| 05    | Method & solver routing       | Select internal SM32 modules and, when required, qualified external tools.                                                |
| 06    | Execution plan                | Build reproducible DAG, parameter sweeps, seeds, manifests, checkpoints and failure semantics.                            |
| 07    | Independent verification      | Require a second numerical/analytical path when risk policy indicates it.                                                 |
| 08    | Validation                    | Compare against benchmark, experimental, reference, or acceptance data.                                                   |
| 09    | UQ & sensitivity              | Quantify numerical/parameter/model uncertainty and sensitivity.                                                           |
| 10    | Discrepancy analysis          | Classify solver disagreement and investigate mesh, model, tolerance, implementation and data causes.                      |
| 11    | Evidence & provenance         | Hash artifacts, track source lineage, limitations, known defects and claim level.                                         |
| 12    | Decision synthesis            | Separate observed result, inference, hypothesis, limitation and recommendation.                                           |
| 13    | Scientific / patent PDF       | Generate evidence-backed scientific report and patent-style technical specification.                                      |

## 7. Claims & Evidence Audit

Claim 1: The system is NOT required to replace every mature engineering, scientific, certification, HPC, CAD, CAE, AI, or regulatory tool.

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

Claim 2: Q0 = UNKNOWN Q1 = AVAILABLE Q2 = TESTED Q3 = BENCHMARKED Q4 = INDUSTRIALLY VALIDATED Q5 = CERTIFICATION-ACCEPTABLE FOR SPECIFIC CONTEXT

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

Claim 3: Always recompute final feasibility independently.

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

Claim 4: IMPLEMENTED TESTED NUMERICALLY\_VERIFIED VALIDATED BENCHMARKED STRESS\_TESTED ENDURANCE\_TESTED EXTERNALLY\_REPRODUCED INDUSTRIALLY\_VALIDATED CERTIFICATION-ACCEPTABLE

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

Claim 5: Never skip directly from IMPLEMENTED to CERTIFICATION-ACCEPTABLE.

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

Claim 6: SUPER MOON's strength must not be pretending to replace every specialized tool.

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

Claim 7: Never claim: CERTIFIED QUALIFIED INDUSTRIAL PARITY PRODUCTION SCALE GPU QUALIFIED MPI QUALIFIED EXTERNALLY REPRODUCED

Attach benchmark, test, qualification, external reproduction, or downgrade claim wording.

## 8. Knowledge Alignment

The following excerpts were retrieved from the integrated SuperMoon knowledge corpus based on the prompt's dominant domains and governance requirements.

### Knowledge hit 1 — lines 6809838–6809838 — SM32Q authoritative successor

Certification acceptance is **not claimed**. MPI multi-node, GPU/multi-GPU runtime qualification, and independent external-machine reproduction are **not executed** in this environment. Longer endurance attempts interrupted by the execution wrapper are not counted.

```
<<<END_SM32Q_FILE>>>
<<<SM32Q_FILE path="evidence/SM32Q_COVERAGE.json"
sha256="23b2757be8d4c5bc96b1d6fcf0aa34428bc226cc0dfe0840325437e64b0dd185" bytes="185100">>>
{"meta": {"format": 3, "version": "7.13.3", "timestamp": "2026-08-19T17:25:29.578985", "branch_coverage": true, "show_contexts": false}, "files":
{"src/supermoon32/qualified/_init_.py": {"executed_lines": [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25,
26, 27, 28], "summary": {"covered_lines": 27, "num_statements": 27, "percent_covered": 100.0, "percent_covered_display": "100",
"missing_lines": 0, "excluded_lines": 0, "percent_statements_covered": 100.0, "percent_statements_covered_display": "100", "num_branches":
0, "num_partial_branches": 0, "covered_branches": 0, "missing_branches": 0, "percent_branches_covered": 100.0,
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"percent_branches_covered": 100.0, "percent_branches_covered_display": "100"}, "missing_lines": [], "excluded_lines": [], "start_line": 1,
"executed_branches": [], "missing_branches": []}}, "src/supermoon32/qualified/change_impact.py"
```

### Knowledge hit 2 — lines 6798293–6798293 — SM32 full implementation successor

```
<<<END_SM32_FILE>>>
<<<SM32_FILE path="evidence/SM32_COVERAGE.json"
sha256="f66a84fe5d7429bc697848058bf2d5de60fdab9722d234248b58e653ded2f9ff" bytes="198188">>>
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{"src/supermoon32/_init_.py": {"executed_lines": [2, 3, 4], "summary": {"covered_lines": 3, "num_statements": 3, "percent_covered": 100.0,
"percent_covered_display": "100", "missing_lines": 0, "excluded_lines": 0, "percent_statements_covered": 100.0,
"percent_statements_covered_display": "100", "num_branches": 0, "num_partial_branches": 0, "covered_branches": 0, "missing_branches": 0,
"percent_branches_covered": 100.0, "percent_branches_covered_display": "100"}, "missing_lines": [], "excluded_lines": [],
"executed_branches": [], "missing_branches": [], "functions": {"": {"executed_lines": [2, 3, 4], "summary": {"covered_lines": 3, "num_statements":
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"executed_branches": [], "missing_branches": []}, "classes": {"": {"executed_lines": [2, 3, 4], "summary": {"covered_lines": 3, "num_statements":
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"percent_statements_covered_display": "100", "num_branches": 30, "num_partial_branches": 1, "covered_branches": 29, "missing_branches":
1, "meta": {"percent_branches_covered": 96.66666666666667, "percent_branches_covered_display": "97"}, "m
```

### Knowledge hit 3 — lines 6812989–6812989 — SM32Q authoritative successor

```
with pytest.raises(InvalidInput):local_sensitivity(lambda x:0,[1],method='bad')

def test_reliability_invalid():
with pytest.raises(InvalidInput):reliability_monte_carlo(lambda x:x,lambda r:1,0)

def test_robust_objective_invalid():
with pytest.raises(InvalidInput):robust_objective(None,[])
```

### Knowledge hit 4 — lines 6812769–6812769 — SM32Q authoritative successor

```
f=lambda x:x[0]**3+np.sin(x[1]);g=complex_step_gradient(f,[2,..3]);assert np.allclose(g,[12,np.cos(.3)],rtol=1e-12)

def test_reliability_mc():
out=reliability_monte_carlo(lambda x:x,lambda rng:rng.normal(),10000,seed=1);assert abs(out['probability_failure']-.5)<.03 and out['n']==10000
```

```
def test_robust_objective():
    assert robust_objective(None,[1,1,1],2)==1
```

### Knowledge hit 5 — lines 6812595–6812595 — SM32Q authoritative successor

```
z=x.astype(complex);z[i]+=1j*h;g[i]=np.imag(fn(z))/h
return g
```

```
def reliability_monte_carlo(limit_state,sampler,n,seed=0):
    if int(n)<=0:raise ValueError('n must be positive')
    rng=np.random.default_rng(seed);fail=0;gvals=[]
    for _ in range(int(n)):
```

### Knowledge hit 6 — lines 6810774–6810774 — SM32Q authoritative successor

```
"lhs_samples",
"local_sensitivity",
"complex_step_gradient",
"reliability_monte_carlo",
"robust_objective"
],
"methods": [],
```

### Knowledge hit 7 — lines 6241350–6245437 — BEGIN SUPER MOON 26.0 COMPILED 150K FEATURE REGISTRY JSONL

...| TARGET=SUPER MOON 25.0 OMEGA RESEARCH / DISCOVERY / SCIENTIFIC COMPUTATION FABRIC | MANDATE: implement reverse-mode AD with tape/graph optimization, checkpointing, custom VJPs, memory controls, and mutation-safe semantics for Symbolic Computation and Computer Algebra inside the Reliability testing formal verification and QA layer, covering unit/property/metamorphic/randomized/differential tests, proof obligations, numerical QA, and regression gates. CROSS-MATH: integrate or explicitly assess Dantzig-Wolfe decomposition where mathematically valid; if invalid, record a typed NOT\_APPLICABLE rationale instead of forcing an analogy. COMPUTE/ARCH: provide an implementation path compatible with MPI-IO plus a portable reference/fallback path; define complexity, memory, precision, determinism, concurrency, and failure semantics. AI/RESEARCH: expose the capability to symbolic, numerical, learning, planning, theorem-proving, program-synthesis, and research-agent workflows through typed interfaces without bypassing correctness checks. VERIFY: gradient checks, memory profiling, higher-order composition tests, and cross-backend consistency; include known-answer, adversarial, scale, and regression cases; emit machine-readable PASS/FAIL/UNKNOWN/NOT\_APPLICABLE evidence and performance metadata. DEPENDENCY\_HINT=F148318; SHARD=M099-L27; PRIORITY=1; PRESERVE the supplied SM25 lineage and existi...

### Knowledge hit 8 — lines 6239128–6241349 — BEGIN SUPER MOON 26.0 150K MASTER PROMPT (VERBATIM)

...immutable IDs, source-to-result trace queries, and tamper-evident manifests; include known-answer, adversarial, scale, and regression cases; emit machine-readable PASS/FAIL/UNKNOWN/NOT\_APPLICABLE evidence and performance metadata. DEPENDENCY\_HINT=F146091; SHARD=M098-L12; PRIORITY=3; PRESERVE the supplied SM25 lineage and existing behavior additively: do not delete capabilities, silently weaken tests, or claim state-of-the-art/replacement parity without workload-specific external evidence. IMPLEMENT real executable code, tests, schemas, documentation, benchmark hooks, and provenance rather than placeholders, TODOs, mocked success, or prose-only stubs. F146093 | M098 Scientific Computing and HPC Mathematics | L12 Machine learning and representation learning | A43 Uncertainty Calibration | TARGET=SUPER MOON 25.0 OMEGA RESEARCH / DISCOVERY / SCIENTIFIC COMPUTATION FABRIC | MANDATE: quantify epistemic/aleatoric/numerical/model uncertainty and calibrate confidence or credible intervals against held-out and synthetic truth for Scientific Computing and HPC Mathematics inside the Machine learning and representation learning layer, covering supervised/self-supervised/contrastive/meta learning, embeddings, kernels, and learned surrogates. CROSS-MATH: integrate or explicitly assess mean-field games where mathematically valid; if invalid, record a typed NOT\_APPLICABLE rationale instead of...

## 9. Recommendations

- Downgrade or evidence 7 high-strength claim(s) before treating the prompt as a scientific conclusion.
- SM32Q escalation is active: internal output should be treated as research/support output until a context-qualified external path is supplied.
- Add a named human review gate and decision authority before final acceptance.

## 10. Reproducibility Record

| Artifact            | Value                                                           |
|---------------------|-----------------------------------------------------------------|
| Prompt SHA-256      | 34f9dce09a261638725cbf6df82a4441814971cfbdabfce4631f618000bb32b |
| Analysis ID         | 34f9dce09a261638                                                |
| Knowledge query     | hypothesis experiment hpc mpi uncertainty monte carlo           |
| Runtime source      | /mnt/data/SUPERMOON_PROMPT_STUDIO/supermoon_runtime/src         |
| External validation | NOT EXECUTED BY THIS REPORT GENERATOR                           |

Appendix A — Master Prompt Excerpt

```
===== BEGIN SUPER MOON 32.0 OMEGA QUALIFIED RESEARCH AUTOMATION VERIFICATION
ORCHESTRATION AUTHORITATIVE SUCCESSOR =====
<<<SM32Q_OUTER_MERGE_RECEIPT>>>
{
"authoritative_runtime": "SM32 + supermoon32.qualified",
"branch_coverage_percent": 98.17708333333333,
"certification_acceptance": "NOT_CLAIMED",
"completed_endurance_seconds": [
10,
15
],
"created_unix": 1787160455.0988,
"format": "SM32Q_RELEASE_RECEIPT_V1",
"overlay_policy": "ALL_PREVIOUS_CODE_RETAINED_BYTE_FOR_BYTE; QUALIFIED_CORE_APPENDED;
LATEST_SM32Q_LAYER_AUTHORITATIVE_FOR_QUALIFIED_ORCHESTRATION",
"previous_gzip_bytes": 87658945,
"previous_gzip_sha256": "831cfccd92167b121b88e658408e2e2bbf6331a269c3cb005d3698dfb9178121",
"previous_member_preserved_byte_for_byte": true,
"qualification_status": "PASS_WITH_LIMITATIONS",
"qualified_payload_manifest_entries": 48,
"qualified_payload_manifest_sha256": "ee522a91fff0dfc67a4e329afea99a7423beea7fc5e173f129daf0a7b54118ca",
"qualified_tests_passed": 130,
"real_external_qualified_tool_execution": "NOT_EXECUTED_NO_TOOL_PROVIDED",
"release": "SUPER MOON 32.0 OMEGA \u2014 2014 QUALIFIED RESEARCH AUTOMATION VERIFICATION ORCHESTRATION CORE",
"sm32q_file_count": 50,
"statement_coverage_percent": 99.58847736625515,
"tests_total_passed": 323
}
<<<END_SM32Q_OUTER_MERGE_RECEIPT>>>
<<<SM32Q_FILE path="README_SM32_QUALIFIED_CORE.md"
sha256="21a41d9df5c1ac3deff2494e6fca38e550afb8c340a1e15e6fd0d1bab966b917" bytes="1230">>>
# SUPER MOON 32.0 OMEGA — Qualified Research, Automation, Verification & Orchestration Core
```

This additive SM32 successor layer implements the qualified research/orchestration master prompt without deleting or rewriting the prior SUPER MOON 32.0 Full Implementation Dominion payload.

The layer adds executable risk-aware computation, mandatory escalation policy, qualified-tool registry and CLI adapter mechanics, solver routing, mathematical problem formalization, dimensional/regime analysis, model-validity checking, independent verification/discrepancy analysis, GCI/Richardson tools, advanced UQ/sensitivity/reliability, adjoint/POD/DMD/Bayesian research math, DOE/surrogates/multi-fidelity helpers, DAG orchestration, provenance/evidence/claim governance, qualification and change-impact ledgers, HPC routing/scaling metrics, multi-solver normalization, co-simulation, precision governance, domain-specific routing policies, benchmark management and endurance management.

Truth policy: this code does **not** claim that any real external CAE/HPC/CAD/ML vendor tool has been certified or executed. External-tool routing mechanics are implemented and tested; real tool qualification remains tool/version/context specific.

```
<<<END_SM32Q_FILE>>>
<<<SM32Q_FILE path="docs/SM32Q_MASTER_PROMPT_EN.txt"
sha256="947983489d19ac80dc918e73ba5d0772fdea9880dd2bb80ff329196afb0cc77" bytes="61257">>>
SUPER MOON — QUALIFIED RESEARCH, AUTOMATION, VERIFICATION & ORCHESTRATION CORE
ADVANCED MATHEMATICS, MULTI-SOLVER GOVERNANCE, RISK-AWARE COMPUTATION,
QUALIFIED-TOOL ESCALATION, VALIDATION, UQ, HPC & EVIDENCE MASTER PROMPT
```

```
=====
0. PRIME DIRECTIVE
=====
```



You are designing the next evolution of SUPER MOON as a high-assurance:

#### RESEARCH + AUTOMATION + VERIFICATION + ORCHESTRATION CORE

The system is NOT required to replace every mature engineering, scientific, certification, HPC, CAD, CAE, AI, or regulatory tool.

Its primary role is to:

1. formulate complex scientific and engineering problems correctly;
2. formalize their governing mathematics;
3. select suitable numerical and computational methods;
4. evaluate risk, criticality, uncertainty, and scale before execution;
5. execute internal SUPER MOON solvers where appropriate;
6. invoke qualified external tools when risk, scale, certification, or safety makes external qualification mandatory;
7. execute independent verification paths;
8. compare solver outputs;
9. quantify uncertainty and numerical error;
10. detect disagreements and investigate their cause;
11. automate parameter studies, design-space exploration, validation, optimization, and reproducibility;
12. produce evidence-backed technical conclusions;
13. maintain strict provenance, auditability, and claim discipline.

Target architecture:

```

PROBLEM
↓
FORMALIZATION
↓
MATHEMATICAL MODEL
↓
RISK / SCALE / CRITICALITY ASSESSMENT
↓
METHOD SELECTION
↓
INTERNAL SOLVER / QUALIFIED EXTERNAL TOOL / BOTH
↓
AUTOMATED EXECUTION
↓
INDEPENDENT VERIFICATION
↓
VALIDATION
↓
UNCERTAINTY QUANTIFICATION
↓
DISCREPANCY ANALYSIS
↓
REPRODUCIBILITY
↓
EVIDENCE PACKAGE
↓
ENGINEERING / RESEARCH DECISION

```

Defining principle:

NO CRITICAL DECISION FROM A SINGLE UNVERIFIED COMPUTATIONAL PATH.

#### =====

#### 1. CORE DESIGN PHILOSOPHY

#### =====

Do not maximize:

- number of algorithms;
- number of classes;
- number of modules;
- number of solver names;
- number of architecture diagrams;

- number of unsupported theoretical claims.

Maximize:

- correctness of problem formulation;
- mathematical rigor;
- solver suitability;
- numerical stability;
- verification depth;
- validation quality;
- uncertainty awareness;
- solver independence;
- reproducibility;
- traceability;
- evidence quality;
- defensibility of decisions.

Quality != FeatureCount

Quality = f(Correctness, Verification, Validation, Uncertainty, Reproducibility, Traceability, Qualification, Evidence)

## 2. THREE-TIER COMPUTATIONAL ARCHITECTURE

### LEVEL A — INTERNAL COMPUTATIONAL CORE

Use SUPER MOON internal implementations for research, prototyping, algorithm development, solver experimentation, comparative studies, low/medium-scale simulations, hypothesis testing, sensitivity analysis, reduced-order models, design-space exploration, surrogate construction, and independent cross-checking.

### LEVEL B — QUALIFIED TOOL ORCHESTRATION

Use externally qualified or mature tools when scale exceeds validated internal capacity; human safety, formal certification, airworthiness, production HPC, industrial CAD robustness, extreme distributed execution, or formal independent authority review requires them.

### LEVEL C — MULTI-SOLVER VERIFICATION

For high-risk problems require combinations such as:

SUPER MOON internal solver

+

qualified solver A

+

qualified solver B

+

analytical / manufactured / experimental reference

Never assume numerical agreement merely because tools are reputable.

## 3. COMPUTATIONAL RISK INDEX

Build a formal Computational Risk Index (CRI).

Define normalized variables in [0,1]:

C = criticality

S = computational scale

U = uncertainty

I = consequence of incorrect result

E = evidence deficiency

N = novelty of modeled regime

Q = qualification deficiency

$$CRI = wC*C + wS*S + wU*U + wI*I + wE*E + wN*N + wQ*Q$$

with  $\text{sum}(\text{weights}) = 1$ .

Suggested classes:

$CRI < 0.20$  LOW

$0.20 \leq CRI < 0.40$  MODERATE

$0.40 \leq CRI < 0.60$  HIGH

$0.60 \leq CRI < 0.80$  VERY\_HIGH

$CRI \geq 0.80$  CRITICAL

Weights must be domain-specific rather than universal constants.

#### 4. MANDATORY ESCALATION CONDITIONS

Qualified external tools become mandatory when any of the following applies:

- human safety;
- life-critical engineering;
- regulatory certification;
- airworthiness approval;
- nuclear safety;
- critical infrastructure;
- medical-device regulatory decisions;
- flight-critical structural verification;
- extreme nonlinear structural analysis;
- very large production CFD;
- billion-cell or comparable distributed CFD;
- extreme sparse distributed systems;
- industrial CAD exchange requiring production-kernel robustness;
- production GPU training beyond validated internal capability;
- formal contractual/legal acceptance criteria;
- independent authority review.

SUPER MOON may still provide research analysis, preprocessing, mesh studies, independent verification, sensitivity analysis, optimization proposals, surrogate models, post-processing, uncertainty analysis, discrepancy analysis, and evidence organization.

But internal output alone must NOT be represented as final authority unless qualification evidence explicitly supports that status.

#### 5. QUALIFIED TOOL REGISTRY

Create QualifiedToolRegistry records containing:

tool\_name  
 vendor\_or\_project  
 version  
 domain  
 solver\_types  
 supported\_equations  
 verified\_use\_cases  
 known\_certification\_contexts  
 supported\_hardware  
 validated\_scale\_range  
 input\_formats  
 output\_formats  
 API  
 CLI  
 automation\_interface  
 license  
 qualification\_level  
 validation\_evidence

benchmark\_evidence  
 known\_limitations  
 known\_failure\_modes  
 supported\_operating\_systems  
 dependency\_requirements  
 reproducibility\_notes

## 6. QUALIFIED TOOL LEVELS

Q0 = UNKNOWN  
 Q1 = AVAILABLE  
 Q2 = TESTED  
 Q3 = BENCHMARKED  
 Q4 = INDUSTRIALLY VALIDATED  
 Q5 = CERTIFICATION-ACCEPTABLE FOR SPECIFIC CONTEXT

Never convert Q1 into Q5 through marketing language.  
 Qualification must remain use-case-specific.

## 7. INTERNAL CAPABILITY MATURITY

M0 = CONCEPT  
 M1 = IMPLEMENTED  
 M2 = UNIT\_TESTED  
 M3 = NUMERICALLY\_VERIFIED  
 M4 = BENCHMARKED  
 M5 = STRESS\_TESTED  
 M6 = EXTERNALLY\_REPRODUCED  
 M7 = INDUSTRIALLY\_VALIDATED  
 M8 = QUALIFIED FOR DEFINED CONTEXT

No internal capability may be silently treated as M8.

## 8. SOLVER ROUTING ENGINE

Create SolverRoutingEngine.

Inputs:

- governing equations;
- dimensionality;
- physics;
- geometry;
- mesh type;
- scale;
- stiffness;
- nonlinearity;
- expected conditioning;
- uncertainty;
- accuracy target;
- runtime budget;
- memory budget;
- hardware;
- criticality;
- certification requirement;
- reproducibility requirement;
- external validation requirement.

Outputs:

- recommended internal solver;
- recommended qualified external solver;

- independent verification solver;
- recommended discretization;
- expected numerical risks;
- validation strategy;
- required evidence;
- escalation decision;
- confidence target.

## 9. SOLVER SELECTION OBJECTIVE

For candidate solver j:

$$J_j = w1 * ExpectedError_j + w2 * Runtime_j + w3 * MemoryCost_j + w4 * QualificationRisk_j + w5 * ModelMismatch_j + w6 * ReproducibilityRisk_j$$

Use Pareto analysis where objectives conflict.

Minimize:

error, runtime, memory, monetary cost, qualification risk, uncertainty, model-form error

subject to:

Accuracy  $\geq$  required\_accuracy  
 Qualification  $\geq$  required\_qualification  
 Memory  $\leq$  available\_memory  
 Runtime  $\leq$  allowed\_runtime  
 Evidence  $\geq$  required\_evidence

## 10. PROBLEM FORMALIZATION ENGINE

No solver may execute before creating:

ProblemDefinition  
 DomainDefinition  
 UnknownVariables  
 GoverningEquations  
 InitialConditions  
 BoundaryConditions  
 MaterialModels  
 ConstitutiveModels  
 Constraints  
 Objectives  
 UncertainParameters  
 Observables  
 ValidationTargets  
 AcceptanceCriteria  
 Units  
 CoordinateSystems  
 ReferenceFrames  
 Assumptions  
 ValidityDomain

## 11. EQUATION DEPENDENCY GRAPH

Represent mathematical dependencies as  $G=(V,E)$ .

Vertices:

variables, fields, equations, constraints, constitutive laws, parameters, solvers, observables, datasets, experiments.

Edges:

dependency, coupling, causality assumption, data flow, discretization dependency, solver dependency.

Use the graph for missing-variable detection, coupling analysis, circular-dependency detection, and verification planning.

## =====

### 12. DIMENSIONAL ANALYSIS ENGINE

## =====

Verify  $[LHS] = [RHS]$  for every equation where dimensional information exists.

Detect:

SI/imperial mismatch  
m/mm mismatch  
Pa/MPa mismatch  
kg/g mismatch  
s/ms mismatch  
rad/degree mismatch  
inconsistent derived units

Abort high-risk workflows on unresolved dimensional inconsistency.

## =====

### 13. NONDIMENSIONALIZATION ENGINE

## =====

Identify useful nondimensional groups where applicable:

Reynolds, Mach, Prandtl, Peclet, Courant, Froude, Knudsen, Biot, Fourier, Strouhal, Rayleigh, Grashof, Weber.

Use them to identify regime, route solvers, assess assumptions, choose discretization, and define stability limits.

## =====

### 14. MODEL REGIME DETECTION

## =====

Detect operating regime rather than blindly applying one model.

CFD:

incompressible, weakly compressible, compressible, transonic, supersonic, hypersonic, laminar, transitional, turbulent, continuum, rarefied.

Structures:

linear, geometrically nonlinear, materially nonlinear, static, transient, modal, buckling, contact-dominated.

Heat transfer:

steady, transient, conduction, convection, radiation, conjugate.

## =====

### 15. MODEL VALIDITY DOMAINS

## =====

Every model must declare:

valid\_parameter\_range  
assumptions  
unsupported\_conditions  
expected\_error\_range  
required\_input\_quality

mesh\_requirements  
time\_step\_requirements  
stability\_constraints  
known\_failure\_modes

Prevent silent extrapolation beyond validity domains.

## =====

### 16. VERIFICATION VS VALIDATION

## =====

VERIFICATION:

Are the equations being solved correctly?

VALIDATION:

Do the equations adequately represent the real system?

Every project must carry independent statuses for both.

## =====

### 17. METHOD OF MANUFACTURED SOLUTIONS

## =====

Choose exact artificial solution  $u^*(x,t)$ , compute  $L(u^*) = f^*$ , solve numerically, and measure  $e_h = ||u_h - u^*||$ .

Report L1, L2, and L-infinity error where appropriate.

Verify expected convergence order.

## =====

### 18. ORDER OF ACCURACY

## =====

For refinement ratio  $r$ :

$$p = \log(e_h/e_{h/r}) / \log(r)$$

Compare observed  $p$  with theoretical order.

Do not call a method second-order based on implementation intent alone.

## =====

### 19. GRID CONVERGENCE INDEX

## =====

Automate coarse/medium/fine solutions.

Record refinement ratio, observed convergence order, asymptotic-range evidence, GCI, and critical observable sensitivity.

## =====

### 20. RICHARDSON EXTRAPOLATION

## =====

Where applicable:

$$\phi_{\text{ext}} = \phi_1 + (\phi_1 - \phi_2)/(r^p - 1)$$

Report assumptions and whether the sequence is in the asymptotic range.

## =====

### 21. INDEPENDENT SOLUTION PATHS

## =====

For important problems create:

PrimarySolution

IndependentVerificationSolution

ReferenceSolution

Independence may come from different discretization, codebase, solver family, mesh, mathematical formulation, analytical reference, or experiment.

## =====

### 22. SOLVER DIVERSITY

## =====

CFD:

internal finite-volume solver + external CFD solver + analytical/MMS reference

FEA:

internal FEM + qualified structural solver + analytical benchmark

Optimization:

internal optimizer + independent optimizer + independent constraint recomputation

Sparse algebra:

internal Krylov + mature reference ecosystem + direct residual verification

## =====

### 23. DISCREPANCY ENGINE

## =====

$\Delta_{AB} = |A-B| / \max(|A|, |B|, \epsilon)$

Classify:

AGREEMENT

MINOR\_DISCREPANCY

SIGNIFICANT\_DISCREPANCY

CRITICAL\_DISCREPANCY

Thresholds must be domain-specific.

## =====

### 24. DISAGREEMENT INVESTIGATION

## =====

Investigate:

geometry, mesh, units, boundary conditions, material properties,

constitutive assumptions, time integ